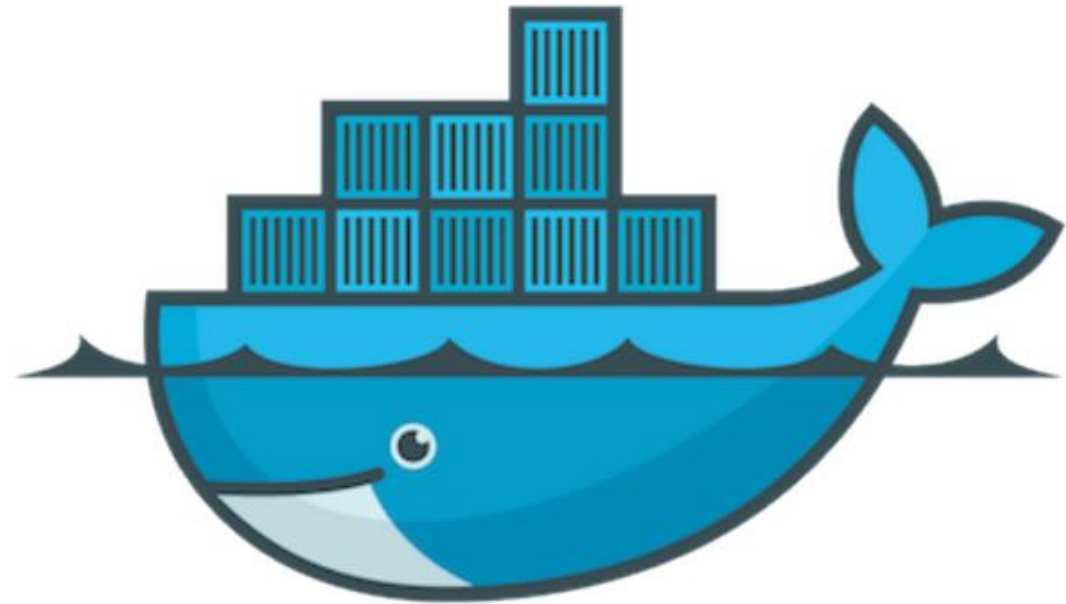


OS 101 FOR DOCKER!



docker

MOHIT VILASRAO DHARMADHIKARI



- I have over 18 years of experience in IT.
- Currently working as *Head Of Engineering* at Merkle.
- Trained over 1000 students in Git!
- Managed git branching strategies at organization level.
- Conducting deep dive sessions on Git since 2021!



BIOS/UEFI

BIOS = Basic Input/Output System

UEFI = Unified Extensible Firmware Interface

POST = Power-On Self-Test

BIOS/UEFI

- ✓ Set the date and time for the hardware clock
- ✓ Disable or enable integrated peripherals
- ✓ Configure error protection
- ✓ Change hardware settings like IRQs and DMA
- ✓ Choose the order of boot devices

BIOS
or
UEFI



Bootloader



Kernel



Init

BIOS



440 bytes
1st device
Bootstrap



512 bytes
MBR
DOS partition
Partition table

POST = Identifies simple hardware failures

- Activates basic components like video output, the keyboard, and storage
- Loads the bootstrap from the MBR
- Loads the second stage of the bootloader and pass options to the kernel

UEFI



NVRAM

EFI applications

FAT filesystems or ISO-9660

EFI System Partition (ESP)

EFI directory

UEFI

POST = Identifies simple hardware failures

- Activates basic components like video output, the keyboard, and storage
- Executes EFI applications stored in the ESP partition, such as the bootloader
- The bootloader loads the kernel
- Also supports Secure Boot

Bootloader



GRUB = Grand Unified Bootloader

Bootloader

acpi	=	Enable or disable ACPI support <code>acpi=off</code> disables
init	=	Set system init <code>Example: init=/bin/bash</code>
systemd.unit	=	Set systemd target <code>Example: systemd.unit=graphical.target</code>
mem	=	Set available system RAM <code>Example: mem=512M</code>
maxcpus	=	Limits processors or cores <code>Example: maxcpus=0</code> or <code>maxcpus=2</code>
quiet	=	Hides boot messages
vga	=	Selects a video mode <code>Example: vga=ask</code>
root	=	Sets the root partition <code>Example: root=/dev/sda3</code>
rootflags	=	Mount options for the root filesystem
ro	=	Mount root filesystem read-only
rw	=	Mount root filesystem read/write

Init

Kernel loaded into RAM

initramfs = initial RAM filesystem

Kernel mounts all filesystems configured in `/etc/fstab`

Kernel loads `init`. First program that starts all others.

initramfs is removed from RAM.

Information and Device Files

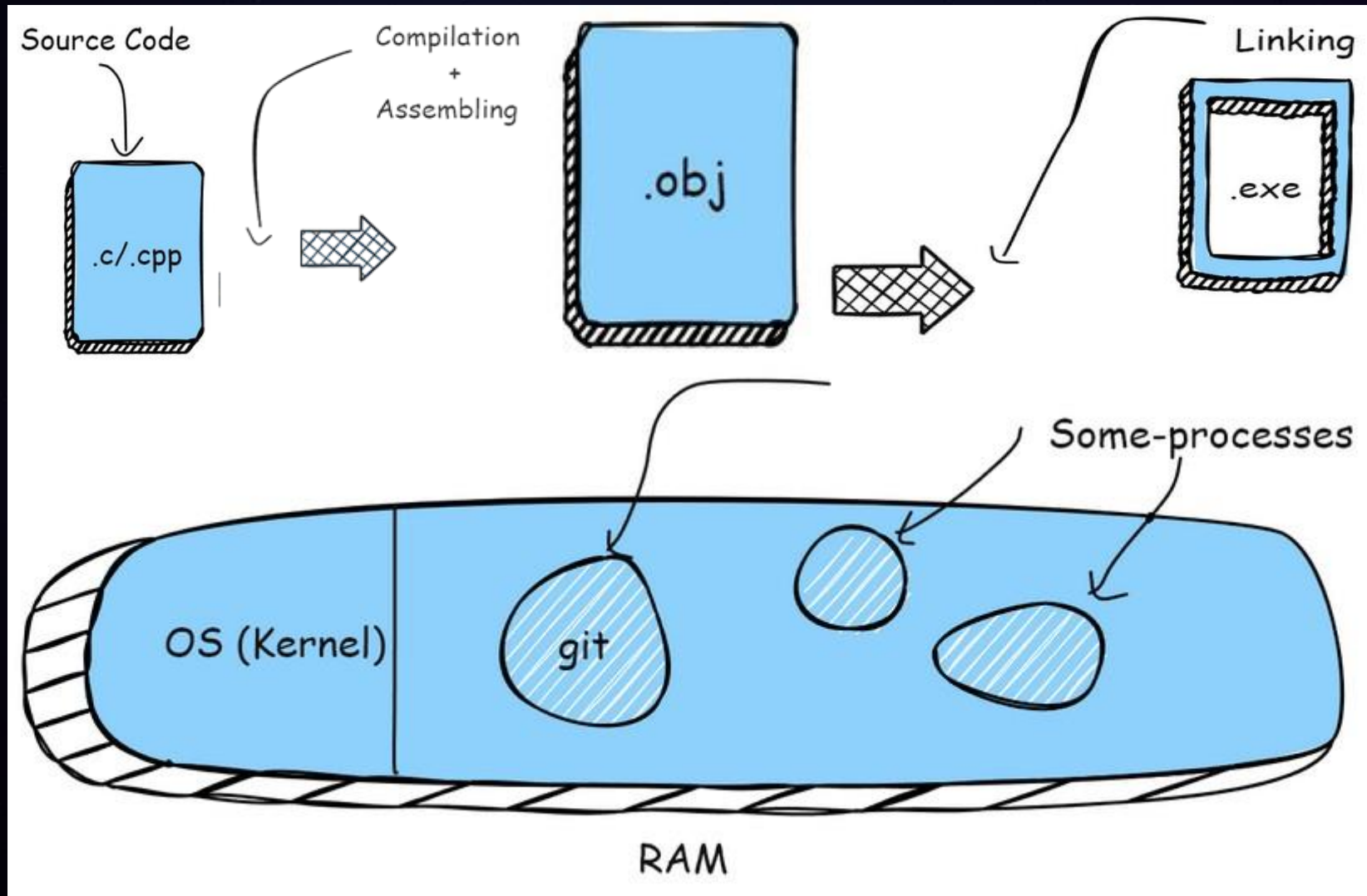
/proc

/sys

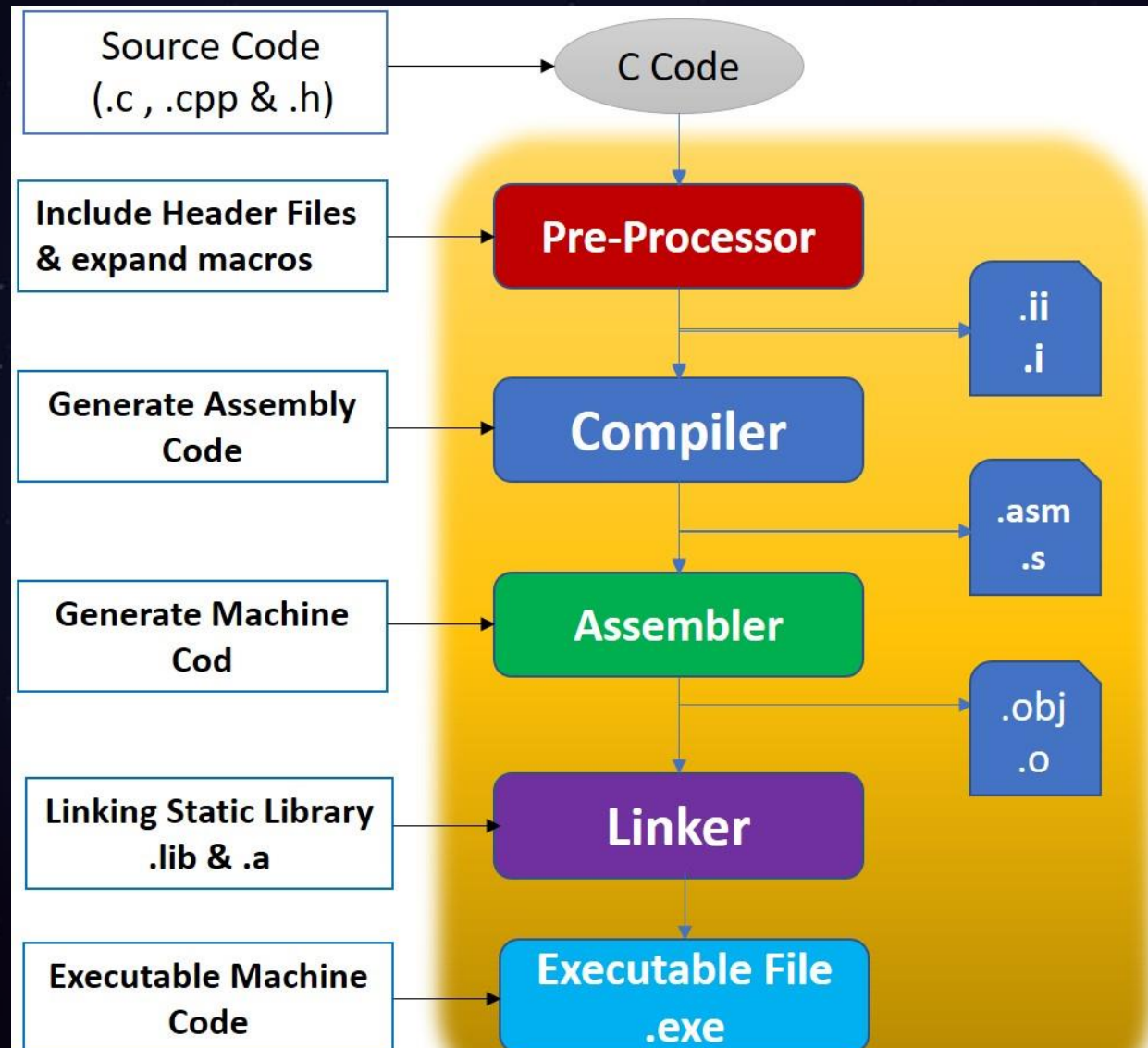
Special directories that exist in RAM

Used by the kernel to store information on running processes

WHAT IS AN APPLICATION



WHAT IS AN APPLICATION



THANK YOU!



Mohit Dharmadhikari

IT Career Mentor & .NET Technical Architect | I help
Aspiring Professionals to grow in Tech by sharing...

